Tensile strength test

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SOURCE: MSC.200(80).pdf
conditioning and must not exceed 5% for al l other specimens. The specimens should show no sign of damage such as shrinking, cracking, swelling, dissolution or change of mechanical qualities.

Tensile strength test

2.6.8 The tensile strength at break of the material should be measured before and after the combined exposure described in 2.6.6.3. When tested according to an international standard acceptable to the Organization , the materials should have a minimum tensile strength of 140 kPa before exposure, which should not be reduced by more than 25% following the combined exposures. In the case of kapok, the protective cover should have a minimum breaking strength of 13 kPa before exposure, which should not be reduced by more than 25% following the combined exposures.

2.7 Donning test

2.7.1 To minimize the risk of incorrect donni ng by uninitiated persons, often in adverse conditions, lifejackets should be examined fo r the following featur es and tested as follows:

.1 fastenings necessary for proper pe rformance should be few and simple, and provide quick and positive closure that does not require tying of knots;

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SOURCE: MSC.200(80).pdf
follows:

.1 fastenings necessary for proper pe rformance should be few and simple, and provide quick and positive closure that does not require tying of knots;

Refer to the recommendations of the International Organization for Standardization, in particular publication ISO 12402-7, Personal flotation devices – Part 7: Materials and components – Safety requirements and test methods.

.2 adult lifejackets should readily fit va rious sizes of adults, both lightly and heavily clad; and

.3 all lifejackets should be capable of being worn inside-out, or clearly in

only one way.

Test subjects

2.7.2 These tests should be carried out with at least 12 able-bodied persons who are completely unfamiliar with the lifejacket an d selected according to the heights and weights in table 2.1 and the following:

.1 small test subjects n eed not be adults;

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SOURCE: MSC.200(80).pdf
completely unfamiliar with the lifejacket an d selected according to the heights and weights in table 2.1 and the following:

.1 small test subjects n eed not be adults;

.2 at least 1/3, but not more than 1/ 2 of test subjects should be females, including at least 1 per height category but excluding the tallest height;

.3 at least one male and one female should be from the lowest and highest weight group;

.4 at least one subject should be selected from each cell containing a “1”; and

.5 enough additional subjects should be selected from cells containing a “X” to total the required number of test subjects, with no more than one subject per cell. A uniform distribution across wei ght ranges should be maintained.

Table 2.1 Test subject selection for adult lifejackets

Weight range – kg
Height
range (m)

40 - 43

43 - 60

60 - 70

70 - 80

80 - 100

100 - 110

110 - 120

>120
< 1.5 1 X X X
1.5 - 1.6 X 1 1 X X

1.6 - 1.7 X X 1 X X
1.7 - 1.8   X X 1 X X X
1.8 - 1.9   X X X 1 1 X
> 1.9   X X X 1

Clothing

2.7.3 Each test subject should be tested wear ing the clothing specified for the test and

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>120
< 1.5 1 X X X
1.5 - 1.6 X 1 1 X X
1.6 - 1.7 X X 1 X X
1.7 - 1.8   X X 1 X X X
1.8 - 1.9   X X X 1 1 X
> 1.9   X X X 1

Clothing

2.7.3 Each test subject should be tested wear ing the clothing specified for the test and appropriate to their size as follows:

.1 Normal clothing means normal indoor clothing, which would not normally interfere with the donning of a lifejacket; and

.2 Heavy-weather clothing means the attire appr opriate for a hostile environment, including a hooded arctic parka and warm cotton gloves.

RESOLUTION MSC.200(80)
(adopted on 13 May 2005)
ADOPTION OF AMENDMENTS TO THE REVISED
RECOMMENDATION ON TESTING OF
LIFE-SAVING APPLIANCESMSC 80/24/Add.1
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2.7.4 Each test should be timed from when the order is given un til the test subject declares that donning is complete. For assessment purposes donning is considered complete when the subject has donned and secu rely adjusted all methods of securing the lifejacket to the extent needed to meet th e in-water performance requirements, including inflation, if needed.

Test without instruction

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SOURCE: MSC.200(80).pdf
complete when the subject has donned and secu rely adjusted all methods of securing the lifejacket to the extent needed to meet th e in-water performance requirements, including inflation, if needed.

Test without instruction

2.7.4.1 The test subjects may be tested indi vidually or as a group. Wearing normal clothing, the first attempt shoul d be with no assistance, guidan ce or prior demonstration. The lifejacket, with closures in the stored condition, should be placed on the floor, face up, in front of the test subject. The instruction provided should be identical for each subject and should be equiva lent to the following: “PLEASE DON THIS LIFEJACKET AS QUICKLY AS POSSIBLE AND ADJUST IT TO A SNUG FIT SO YOU CAN ABANDON SHIP.” The lifejacket should be capable of being donned by at least 75 % of the subjects, and within 1 min. If a subject dons the lifejacket substantially correctly but fails to secure and/or adjust all closures, th e jump test in 2.8.8 a nd in-water performance tests in 2.8.5 and 2.8.6 should be performe d with the lifejacket as donned to establish whether the performance is acceptable and the donning is successful.

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SOURCE: MSC.200(80).pdf
tests in 2.8.5 and 2.8.6 should be performe d with the lifejacket as donned to establish whether the performance is acceptable and the donning is successful.

Test after instruction